

N200 VIO Module User Manual V2.1

This document describes how to use the N200 Visual-Inertial Odometry (VIO) module in a flight controller.

1. Basic Principles

The N200 module estimates the drone's motion relative to the ground by observing ground textures through a downward-facing camera. This enables reliable positioning when GPS is unavailable or denied.

The module does not require preloaded satellite maps and does not rely on matching against existing maps. It can operate directly in any unfamiliar area as long as there is ground texture.

The module uses a visible-light camera and a laser rangefinder to observe the ground. Combined with the drone's current attitude, it outputs both positioning information and terrain height to the flight controller via a serial interface. Under normal conditions, the module outputs positioning data at 30 Hz. When operating over areas with no texture or very weak texture, the module cannot provide valid positioning data and will temporarily stop outputting positioning information. Once valid texture is detected again, positioning output resumes automatically.

Because heading information must be provided by the magnetometer in GPS-denied conditions, magnetometer performance directly affects VIO accuracy. It is recommended to use a high-precision, interference-resistant external magnetometer on your drone.

Note: The visual module provides only positional data. The flight controller still requires heading and altitude information. Please ensure your aircraft is properly configured with a compass, barometer, or other equivalent sensors.

2. Installation of the Module

2.1. Installation Requirements

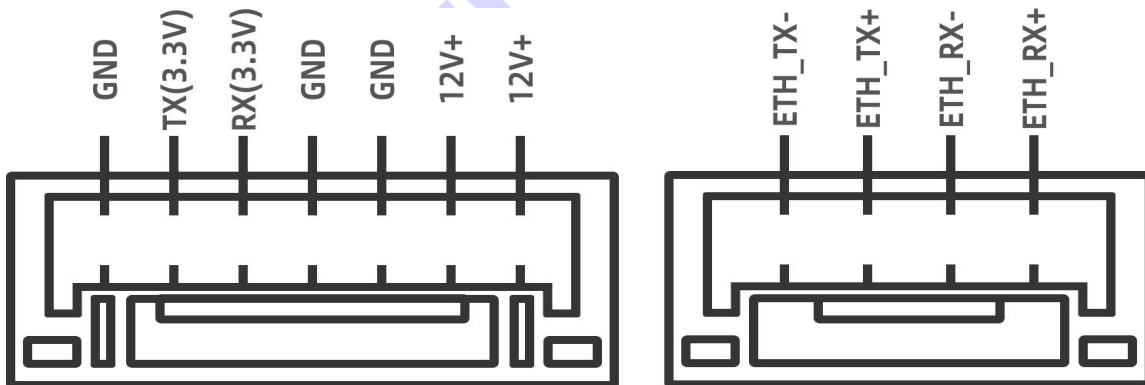
Mount the module underneath the airframe as shown, ensuring that the downward-facing camera points vertically toward the ground and is positioned as close as possible to the flight controller's center. The arrow on the module should point toward the nose of the drone. If the module cannot be aligned exactly with the nose, the orientation must be configured on the parameter configuration page.



2.2. Interface Definition

The module has two interfaces:

- **UART interface (with power):** Used for power supply and communication with the flight controller.
- **Ethernet interface:** Used for connecting to a PC for debugging and configuration. This interface is generally not connected during normal operation.



2.3. Indicator Light

- **Solid on:** Flight controller connection is normal; the module is operating properly.
- **Fast flashing:** Abnormal communication with the flight controller.
- **Slow flashing:** Module malfunction.
- **Off:** No power supply.

3. Module Usage

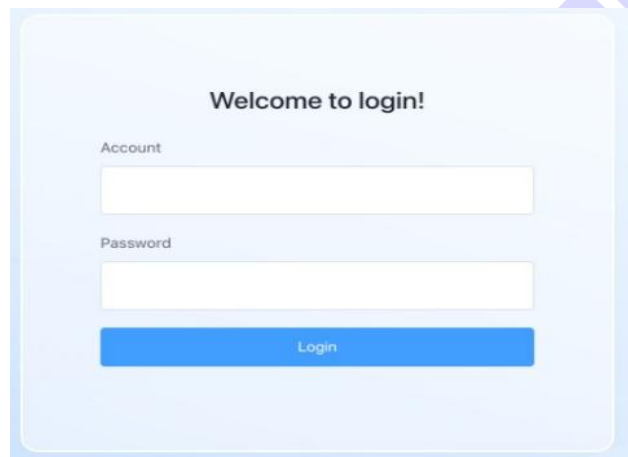
The module supports **PX4** and **ArduPilot** flight controllers. The default mode is PX4. If want to perform functional configuration of the module and upgrade it, use the configuration page.

Connect the module's Ethernet port to a PC and ensure the PC is on the same subnet. Open a web browser and enter:

http://192.168.144.8

Login credentials:

- **Username:** user
- **Password:** 123456

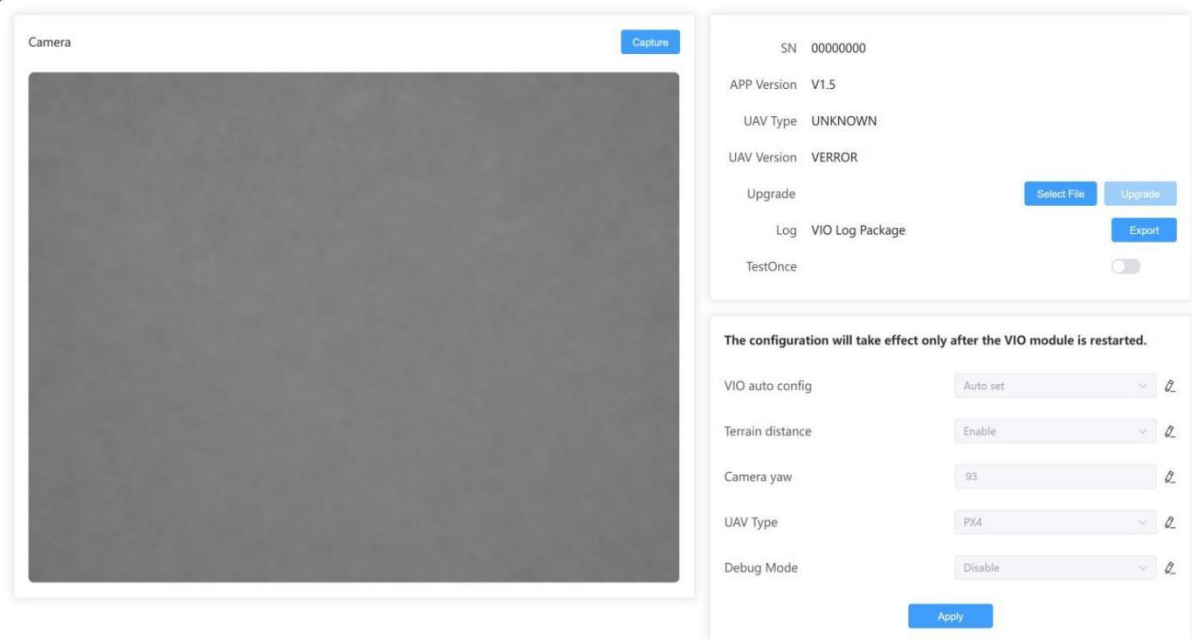


Welcome to login!

Account

Password

Login



Camera

Capture

SN 00000000

APP Version V1.5

UAV Type UNKNOWN

UAV Version VERROR

Upgrade

Log VIO Log Package

TestOnce

Select File Upgrade Export

The configuration will take effect only after the VIO module is restarted.

VIO auto config Auto set

Terrain distance Enable

Camera yaw 03

UAV Type PX4

Debug Mode Disable

Apply

From this page, you can upgrade the VIO module. Click **Select File**, choose the installation package from your local directory, then click **Upgrade**.

To avoid positioning drift caused by unstable vision near the ground, visual positioning starts only when the drone is higher than 5 meters . For easier integration and testing, a TestOnce option is available on the configuration page. When enabled, the 5-meter limit is disabled, allowing visual positioning output even at low altitude. After each reboot, this option is disabled by default.

3.1. PX4

The module currently supports **PX4 v1.15** and **v1.16**.

3.1.1. Connection Setup

Assume the module is connected to **serial port 3** of the flight controller, and take a available MAVLink (e.g. instance 2) as an example to show how to connect.

In QGroundControl parameter settings or the NSH :

1. Set **MAV_2_CONFIG = TELEM 3**
2. Reboot the flight controller
3. Set **SER_TEL3_BAUD = 921600 8N1**
4. Reboot both the flight controller and the VIO module

If a different serial port is used, adjust the configuration accordingly.

You can verify correct operation using the following methods:

1. Observe the indicator light; For details, refer to the indicator light description.
2. Check MAVLink status in the MAVLink console to confirm MAVLink configuration.
3. Check whether the flight controller firmware version is correctly read on the Ethernet configuration page, indicating normal communication between the module and the flight controller.
4. Execute **listener vehicle_visual_odometry** in the MAVLink console to view the transmission frequency and data values sent by the VIO module, indicating that the VIO module is operating correctly. Use the **TestOnce** function if necessary.

3.1.2. VIO-Side Configuration (PX4)

Under normal conditions, keep the parameters on the webpage set to their default values. For advanced control, the following parameters are available.

VIO Automatic Configuration

In PX4, the **EKF2_EV_CTRL** parameter determines how VIO output.

The way **VISION_POSITION_ESTIMATE** participates in flight controller state estimation fusion: **EKF2_EV_DELAY** configures data delay compensation, and **EKF2_EV_NOISE_MD** configures the noise model.

Considering that some drone require enabling or disabling the VIO module based on GPS reliability—specifically, activating the VIO module only when GPS fails and deactivating it when GPS is operational—an option is provided to determine whether this module should automatically configure the PX4 flight controller. The available parameters are as follows.

Note: **EKF2_EV_DELAY** only takes effect after rebooting the flight controller.

Options:

- **0 – Never set:** The module does not modify **EKF2_EV_CTRL**, **EKF2_EV_DELAY**, or **EKF2_EV_NOISE_MD**, but still outputs **VISION_POSITION_ESTIMATE**. Parameter management is handled entirely by the flight controller.
- **1 – Automatic set:** On startup, the module automatically modifies relevant parameters based on the PX4 firmware version.

Default is **1 – Automatic set**. When the flight controller manages parameter configuration on its own, all of the above parameters must be set manually.

Terrain Height

During position estimation, terrain height information is required from PX4. The module includes a downward-facing laser rangefinder to provide distance-to-ground data. PX4 filters this data and feeds it back to the VIO system. If the flight controller wants to use another terrain height sensor, this function should be disabled.

0 – Disabled: When disabled, the **EKF2_TERR_MASK** parameter is not set, and no **DISTANCE_SENSOR** data is sent to the flight controller.

1 – Enabled: When enabled, the **EKF2_TERR_MASK** parameter is set to allow PX4 terrain estimation to fuse rangefinder data, and **DISTANCE_SENSOR** data is sent to the flight controller so that terrain height can be fused. To reduce the impact of terrain height fusion on the drone's altitude estimation: in **PX4 v1.15** the parameter **EKF2_RNG_CTRL** is set to 0, disabling the use of **DISTANCE_SENSOR** data in altitude estimation. In **PX4 v1.16**, due to changes in the flight controller's state estimation, **EKF2_RNG_CTRL** must be set to 1. To minimize its effect on altitude estimation, the **EKF2_RNG_A_HMAX** parameter must be configured manually.

The default setting is 1 – Enabled.

Camera Extrinsic

Due to installation tolerances, the module may not be perfectly aligned with the drone's nose, resulting in a yaw offset. In such cases, the offset must be corrected using the **camera_yaw** parameter.

With the drone nose pointing forward and viewed from above, if the module is rotated to the left relative to the nose, **camera_yaw** should be negative; if the module is rotated to the right, **camera_yaw** should be positive. The value represents the specific yaw offset in degrees (°)

3.1.3. Flight Controller Configuration (PX4)

To improve stability when using the VIO module, the following parameter adjustments are recommended:

Parameter	Default	Recommended
COM_POS_FS_DELAY	1	5
COM_POS_FS_EPH	5	-1

COM_POS_FS_DELAY

When flying close to the ground or over weak/no-texture areas, VIO may temporarily fail. If no other positioning sensors are available, the drone may enter failsafe or degraded modes, potentially causing takeoff failure or forced landing. **COM_POS_FS_DELAY** controls the delay before PX4 triggers position-loss failsafe. The default value of 1 second is often insufficient; **5 seconds** is recommended.

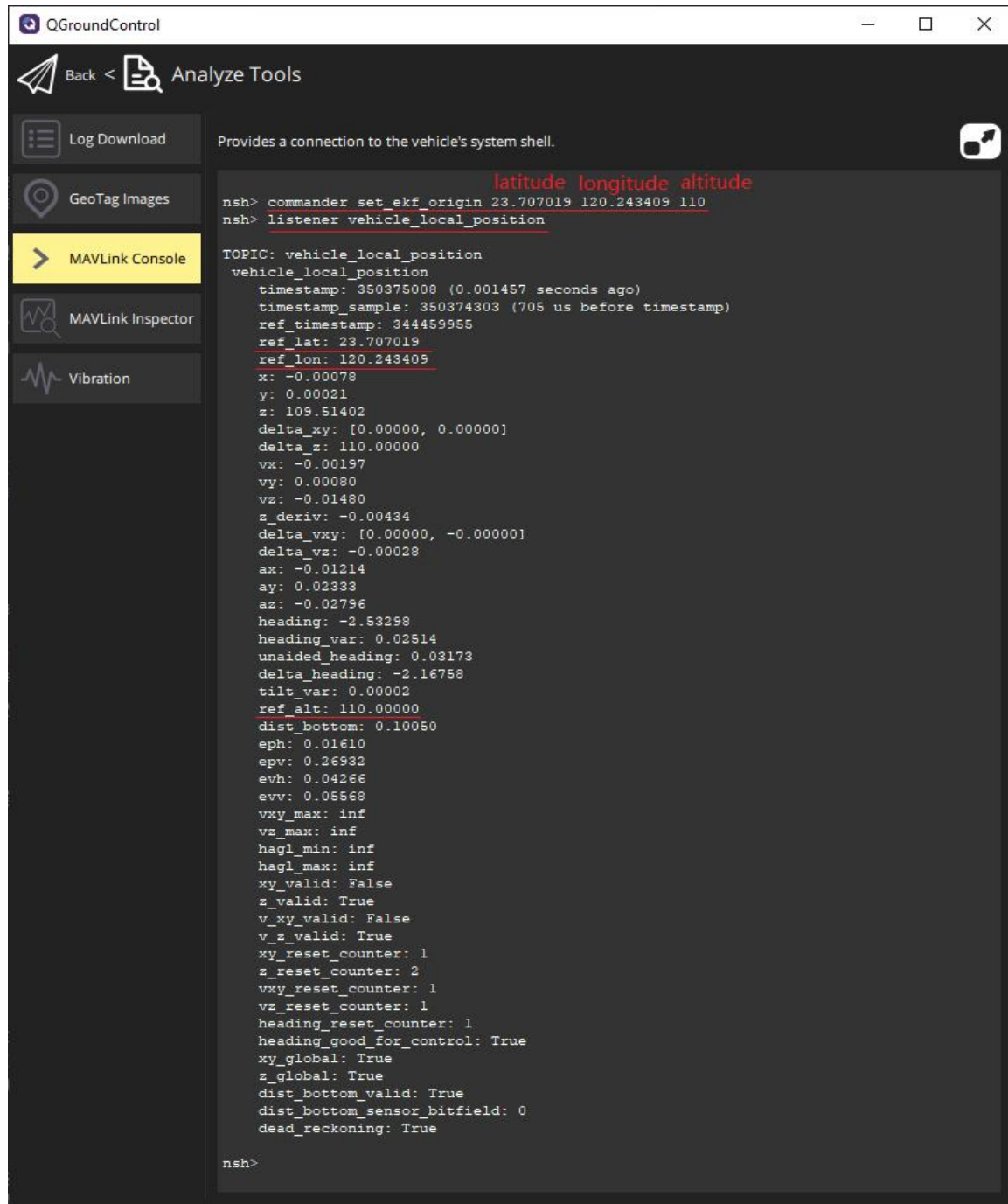
COM_POS_FS_EPH

Horizontal position error threshold, PX4 uses the **COM_POS_FS_EPH** parameter to determine whether the position error exceeds the threshold. If the threshold is exceeded, the position is considered invalid and failsafe or degradation logic is triggered. The default value is 5; it is recommended to set it to -1, meaning that position validity is not judged based on position error.

3.1.4. GPS-Denied Flight (PX4)

If no GPS sensor is present, the drone does not know its absolute global position. Since the module only provides relative positioning from the takeoff point, the true latitude, longitude, and altitude of the takeoff location must be set manually to enable mission flight.

In QGroundControl → Analyze Tools → MAVLink Console, enter: **commander set_ekf_origin lat lon alt** To verify, run: **listener vehicle_local_position**



The screenshot shows the QGroundControl interface with the MAVLink Console open. The console displays the output of the `commander set_ekf_origin 23.707019 120.243409 110` command, followed by the `listener vehicle_local_position` command. The output shows the vehicle's local position and various sensor data.

```
nsh> commander set_ekf_origin 23.707019 120.243409 110
nsh> listener vehicle_local_position

TOPIC: vehicle_local_position
vehicle_local_position
timestamp: 350375008 (0.001457 seconds ago)
timestamp_sample: 350374303 (705 us before timestamp)
ref_timestamp: 344459955
ref_lat: 23.707019
ref_lon: 120.243409
x: -0.00078
y: 0.00021
z: 109.51402
delta_xy: [0.00000, 0.00000]
delta_z: 110.00000
vx: -0.00197
vy: 0.00080
vz: -0.01480
z_deriv: -0.00434
delta_vxy: [0.00000, -0.00000]
delta_vz: -0.00028
ax: -0.01214
ay: 0.02333
az: -0.02796
heading: -2.53298
heading_var: 0.02514
unaided_heading: 0.03173
delta_heading: -2.16758
tilt_var: 0.00002
ref_alt: 110.00000
dist_bottom: 0.10050
epf: 0.01610
epv: 0.26932
evh: 0.04266
evv: 0.05568
vxy_max: inf
vz_max: inf
hagl_min: inf
hagl_max: inf
xy_valid: False
z_valid: True
v_xy_valid: False
v_z_valid: True
xy_reset_counter: 1
z_reset_counter: 2
vxy_reset_counter: 1
vz_reset_counter: 1
heading_reset_counter: 1
heading_good_for_control: True
xy_global: True
z_global: True
dist_bottom_valid: True
dist_bottom_sensor_bitfield: 0
dead_reckoning: True

nsh>
```

3.1.5. Troubleshooting

Issue	Check
No ODOMETRY messages	Serial baud rate / wiring / module power status
EKF not fusing vision	EKF2_EV_CTRL misconfiguration / incorrect output frequency / timestamp issues
Excessive drift	Camera blur / insufficient ground texture

3.2. ArduPilot

The module currently supports **Copter v4.6.0**. Parameters may differ in other versions.

3.2.1. Connection Setup

Assume the module is connected to a serial port on the flight controller (e.g. SERIAL2). In Mission Planner or QGroundControl:

- Set SERIAL2_BAUD = 921600
- Set SERIAL2_PROTOCOL = 1 (MAVLink)
- Reboot the flight controller and the module

If another serial port is used, simply replace it with the corresponding serial port number. After the configuration is completed, verify whether it is correct by observing the indicator status. Refer to the indicator light description for details.

3.2.2. VIO-Side Configuration (ArduPilot)

VIO Automatic Configuration

In ArduPilot, VIO fusion is controlled mainly by VISO_* parameters:

- **VISO_TYPE** = 1: Selects whether to use external visual odometry and specifies its source type.
- **VISO_DELAY_MS** = 40: Sets the delay compensation of visual data relative to the flight controller's internal time base (in milliseconds).
- **VISO_POS_M_NSE** = 10: Sets the noise level of visual position observations (in meters).
- **FS_EKF_THRESH** = 1: EKF failsafe trigger threshold

Options:

- **0 – Never set:** The module does not configure the parameters **VISO_TYPE**, **VISO_DELAY_MS**, **VISO_POS_M_NSE**, or **FS_EKF_THRESH**, but it will still transmit **VISION_POSITION_ESTIMATE** data. Parameter management is handled by the flight controller.
- **1 – Automatic set:** The module automatically modifies parameters based on the ArduPilot firmware version.

By default, this parameter is set to 1 (Automatic configuration). When the flight controller manages parameter settings on its own, all of the above parameters must be configured manually

Terrain Height (ArduPilot)

This parameter is not used in ArduPilot.

3.2.3. Flight Controller Configuration (ArduPilot)

To enable the VIO module in ArduPilot:

1. Enable EKF3 by setting **EK3_ENABLE** = Enabled
2. Set the position source of a free EKF3 instance to **ExternalNav** (e.g. change **EK3_SRC_POSXY** from None to ExternalNav)
3. Reboot the flight controller

If visual positioning is used, the EKF estimator must be switched to the corresponding source. The specific switching method can be found at:

<https://ardupilot.org/copter/docs/common-ekf-sources.html>

To view visual positioning data from the flight controller indoors, the **TestOnce** function must be enabled.

3.2.4. GPS-Denied Flight (ArduPilot)

If no GPS sensor is present, the drone does not know its current position. Since the module only provides relative positioning from the takeoff point, the true latitude, longitude, and altitude of the takeoff location must be set manually to enable mission flight.

In MAVProxy, set the takeoff origin: `setorigin lat lon atl`

```
MAVProxy
Warning: multiple possible serial ports. Use console GUI or 'link add' to add port, or restart using --master to select a single port
Loaded module graph
Loaded module console
PyInstaller\loader\pyimod02_importers.py:419: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setuptools<81.
PyInstaller\loader\pyimod02_importers.py:419: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setuptools<81.
Loaded module map
Running script (C:\Users\win11\AppData\Local\.mavproxy\mavinit.scr)
Loaded module help
Log Directory:
Telemetry log: mav.tlog
Adding link COM7:115200
Connect COM7 source_system=255
Detected vehicle 1:1 on link 0
Received 1048 parameters (ftp)
Saved 1048 parameters to mav.parm
STABILIZE>
STABILIZE> latitude longitude altitude
STABILIZE> setorigin 23.707019 120.243409 110
Setting origin to: 23.707019 120.243409 110.0
STABILIZE>
STABILIZE>
STABILIZE>
STABILIZE>
STABILIZE>
STABILIZE>
```